Objectives

1. Basic concepts of recognizing Pediatric Immune Deficiency Disorders (PIDD)
2. Common immunodeficiencies
   - Phagocyte disorder: Lymphocyte Adhesion Deficiency (LAD)
   - Combined T and B disorder: Severe Combined Immune Deficiency (SCID)
   - T cell disorder: DiGeorge Syndrome
   - B cell disorder: Bruton X linked Agammaglobinemia, Hyper IgM, Job syndrome
   - Other: Wiskott-Aldrich Syndrome (WAS), Ataxia-telangiectasia (AT)
3. Resources
4. Summarization

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Starter Question

- A 2 month old M infant is brought into your office with a recent hospital admission for 2 weeks for Pneumocystis (PCP) Pneumonia requiring respiratory support, in reviewing the chart, the infant has had multiple episodes of thrush that required prolonged antifungal treatment. Growth chart shows the infants head circumference, weight and length are below the <1.5%tile, on exam, he has severe seborrheic dermatitis and a bulging R TM. You review the xray from the hospital and they comment on an absent thymic shadow. Flow Cytometry shows decreased B, T and NK cells and decreased immunoglobulins. What is the most likely mechanism causing this presentation?

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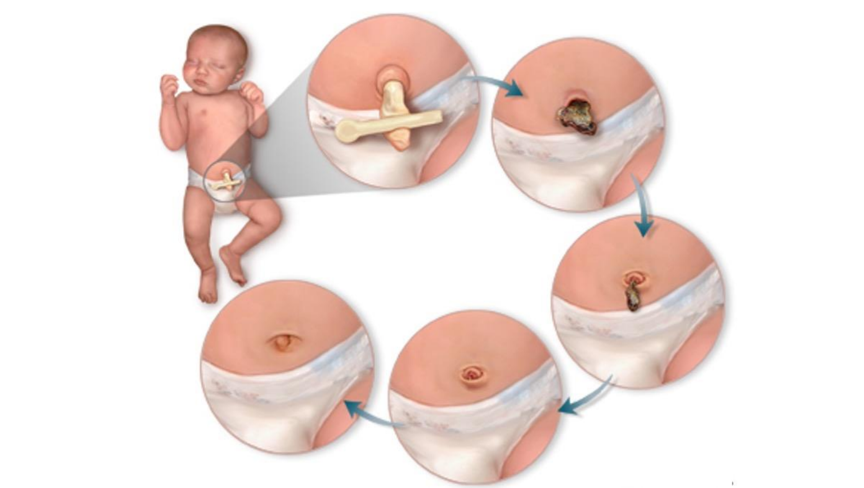
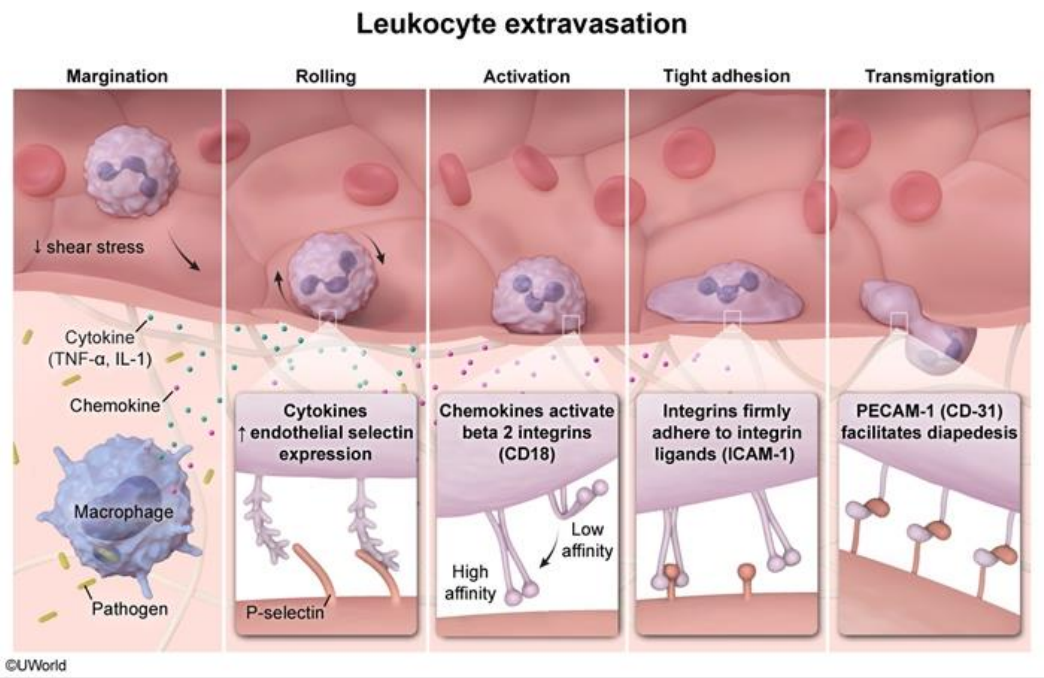
Primary immune deficiency disorders (PIDD)

- 550 and counting different disorders
- Diagnosis tough because
  - Low index of suspicion
  - Low disease incidence
  - Infections, cutaneous lesions, failure to thrive, autoimmunity (can present similar to non specific viral illnesses, or allergies etc) require coordination of care
  - Varied presentations
  - Complex phenotypes
  - Preschool children normally get 6-10 viral infections a year
- Exceed this amt, or severity of symptoms or very prolonged duration – indicating trouble clearing pathogen
- Usually requiring hospitalization for severe symptoms
- Prolonged or ineffective IV abx
- Unusual pathogens (opportunistic infections) or infections from live attenuated vax (rotavirus, VZV, MMR)
- Neutropenia/Lymphocytopenia may be present
- Family hx of PIDD or consanguinity
- Immune dysregulation: autoimmune disease or autoinflammatory conditions (early onset SLE, very early onset IBD, fever syndromes)

Phagocyte disorder: Leukocyte Adhesion Deficiency (type 1)

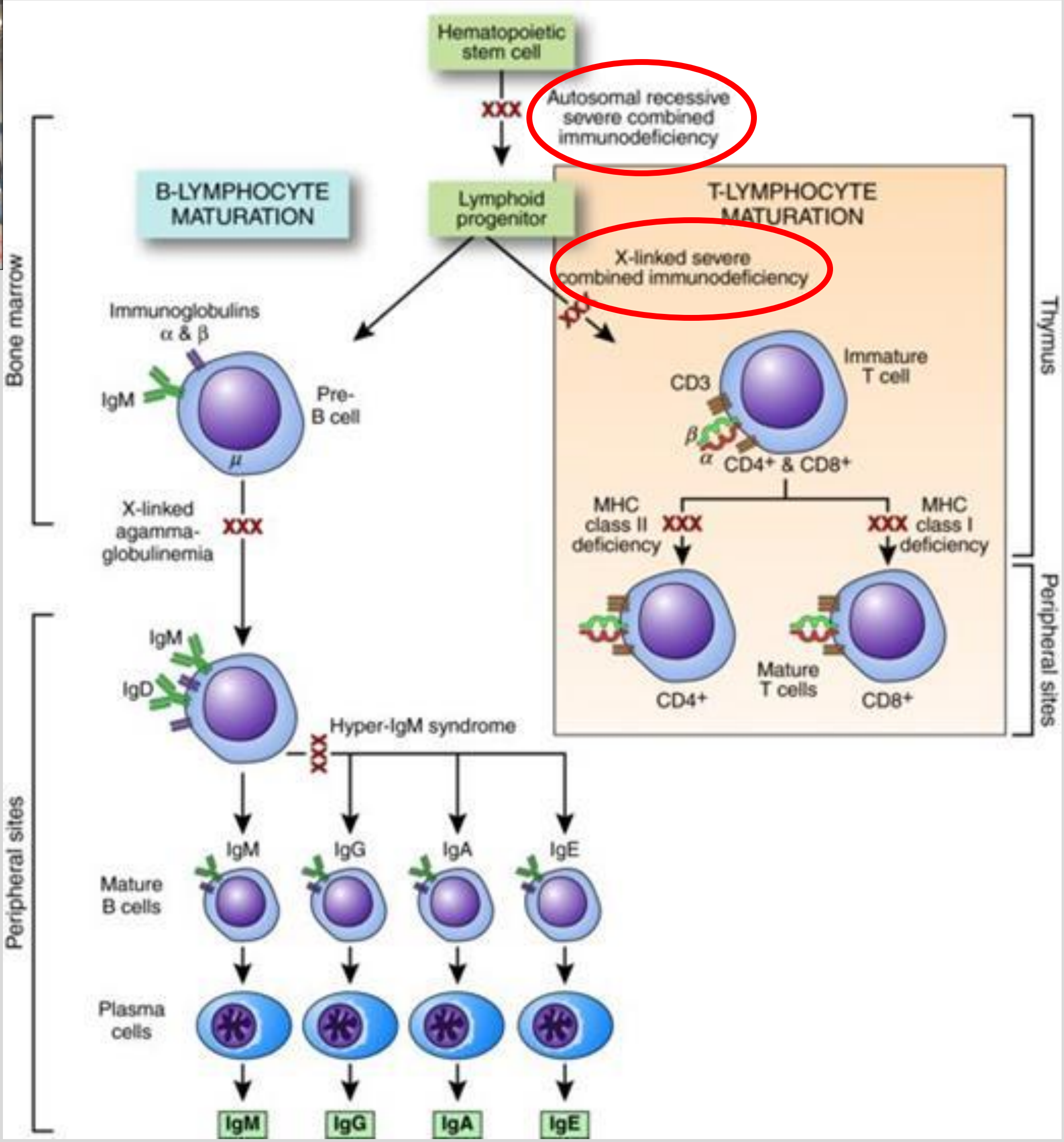
- Autosomal recessive
- Defect in beta-2 integrin (CD18) protein
  - Also called Leukocyte function-associated antigen type 1 (LFA-1)
- Mostly affecting Phagocytes and chemotaxis and migration to site of need
- Clinically:
  - Recurrent skin and mucosal bacterial infections
  - Absent PUS (phagocytic cells can't get to area of need causing inflammation, walling off etc)
  - Impaired wound healing
  - Delayed separation of umbilical cord (normally 14 days, delayed >30 days)
  - Risk for Omphalitis and sepsis
  - Severe gingivitis
- Labs:
  - Neutrophilia >20k
  - Absence of neutrophils at infection site
- Treatment:
  - Ppx Abx
  - Stem cell transplant





Combined T and B cell disorder: Severe combined immunodeficiency (SCID)

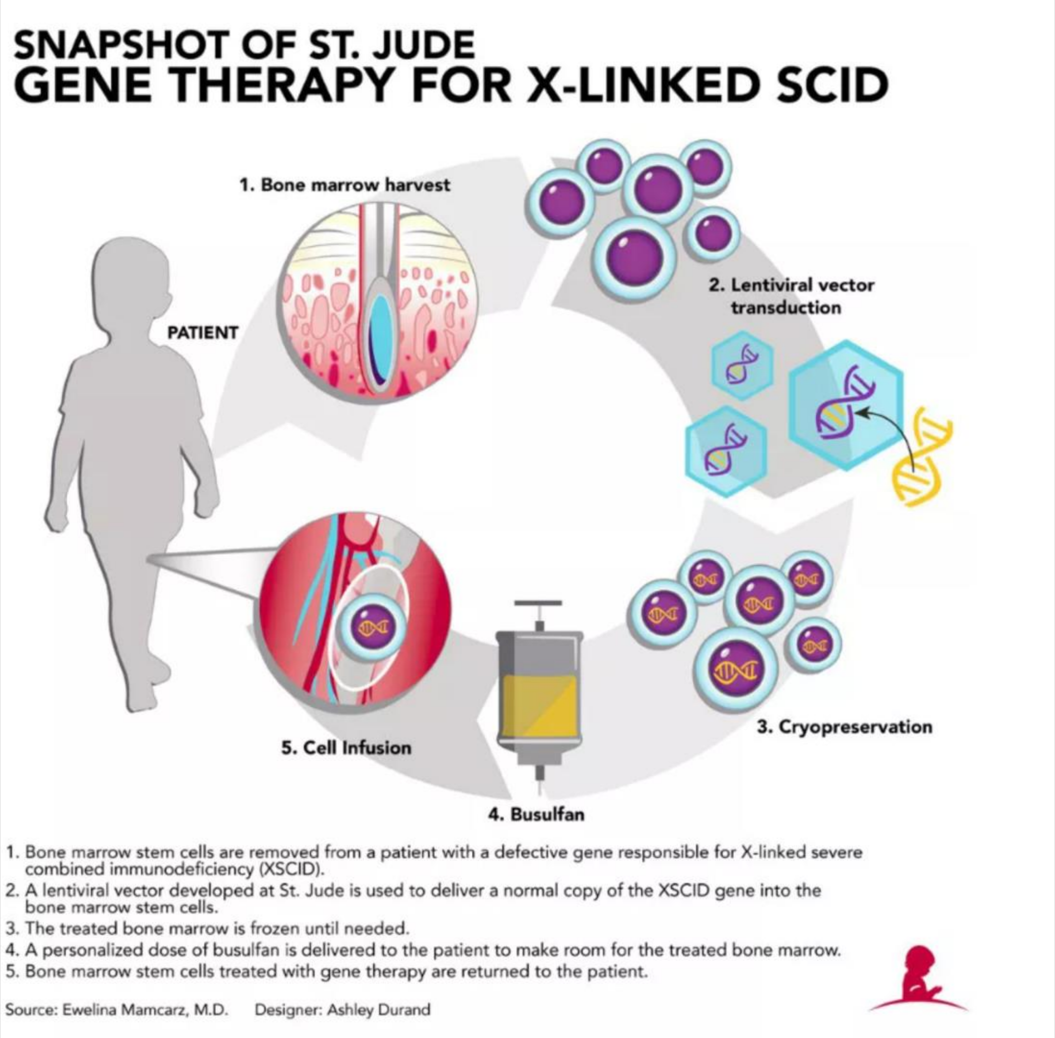
- Bubble baby disease
- Variants of 20 genes involved
  - Defective IL-2R gamma chain (most common) X linked recessive
  - Adenosine Deaminase Deficiency – autosomal recessive
- Presents within first month – 2 months of life
- Loss of cell mediated AND humoral immunity
- Absence of thymic shadow, germinal centers
- T cells absent, very low/undetectable T cell receptor excision circles (TRECs) * on newborn screen
- Diagnosis: flow cytometry – no T cells, B cells can be present but non functional (in X linked), in ADA T, B and NK cells down.
  - In ADA: increased urine deoxyadenosine
- Failure to thrive, chronic diarrhea, thrush
- Recurrent bacterial, fungal, viral, protozoal infections



Combined T and B cell disorder: Severe combined immunodeficiency (SCID)

Treatment:

- Abx, antiviral, antifungal ppx
- Immunoglobin replacement therapies
- Avoid live attenuated vaccines
- Curative:
  - Hematopoietic stem cell transplant
    - Involves conditioning with chemo drugs– makes space for treated stem cells in bone marrow, prevent rejection and incomplete graft
    - May chose not to if fully matched sibling donor or immune system very down or baby <2 months
  - For ADA deficient: enzyme replacement therapy which then takes over function to break down harmful substance
    - Weekly IM injections
    - Temporary
  - Gene therapy
    - Can get away with low dose conditioning
    - Not FDA approved yet – clinical trials going on



T cell disorder: Thymic aplasia (DiGeorge Syndrome)

- 22q11.2 microdeletion
- Failure of development - 3rd and 4th pharyngeal pouches
  - Absent thymus and parathyroid (immune dysfunction and hypocalcemia)
  - Cleft palate + cleft lip
  - Facial abnormalities – micrognathia
  - Cardiac Defects (tetralogy of Fallot, truncus arteriosus)
- T cells don't get educated, decreased #s in periphery (but B and NK cell #s are fine)
  - Paracortex in lymph nodes will be underdeveloped
- Recurrent viral/fungal infections
- Absent thymic shadow on CXR in neonates
- Treatment:
  - Temporary: ppx antivirals, antibiotics, antifungals
  - One time thymic tissue transplant – FDA approved in 2021

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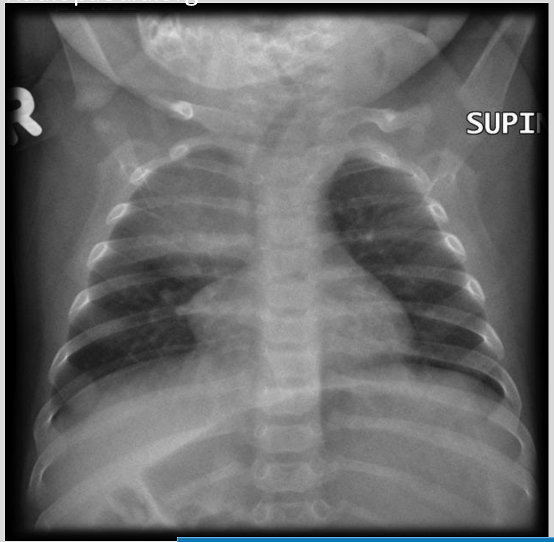
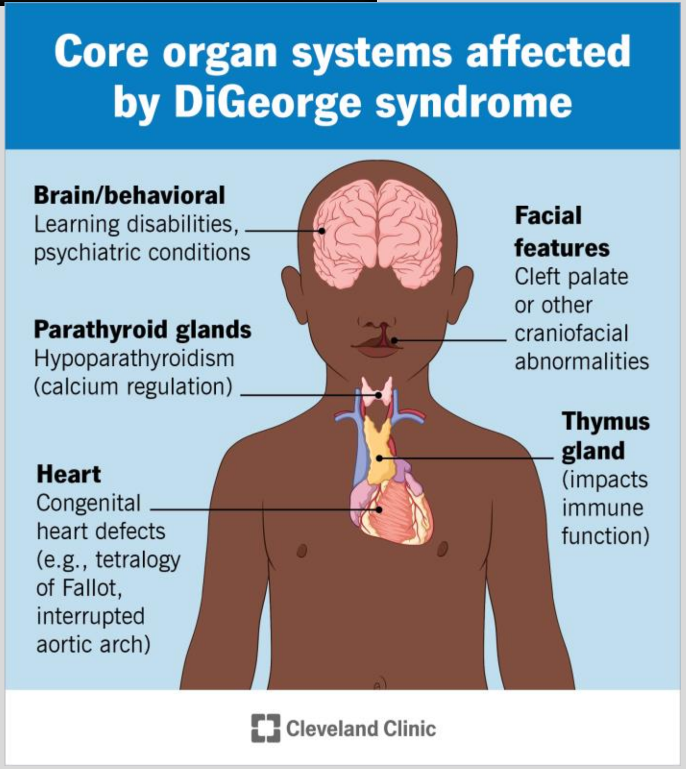
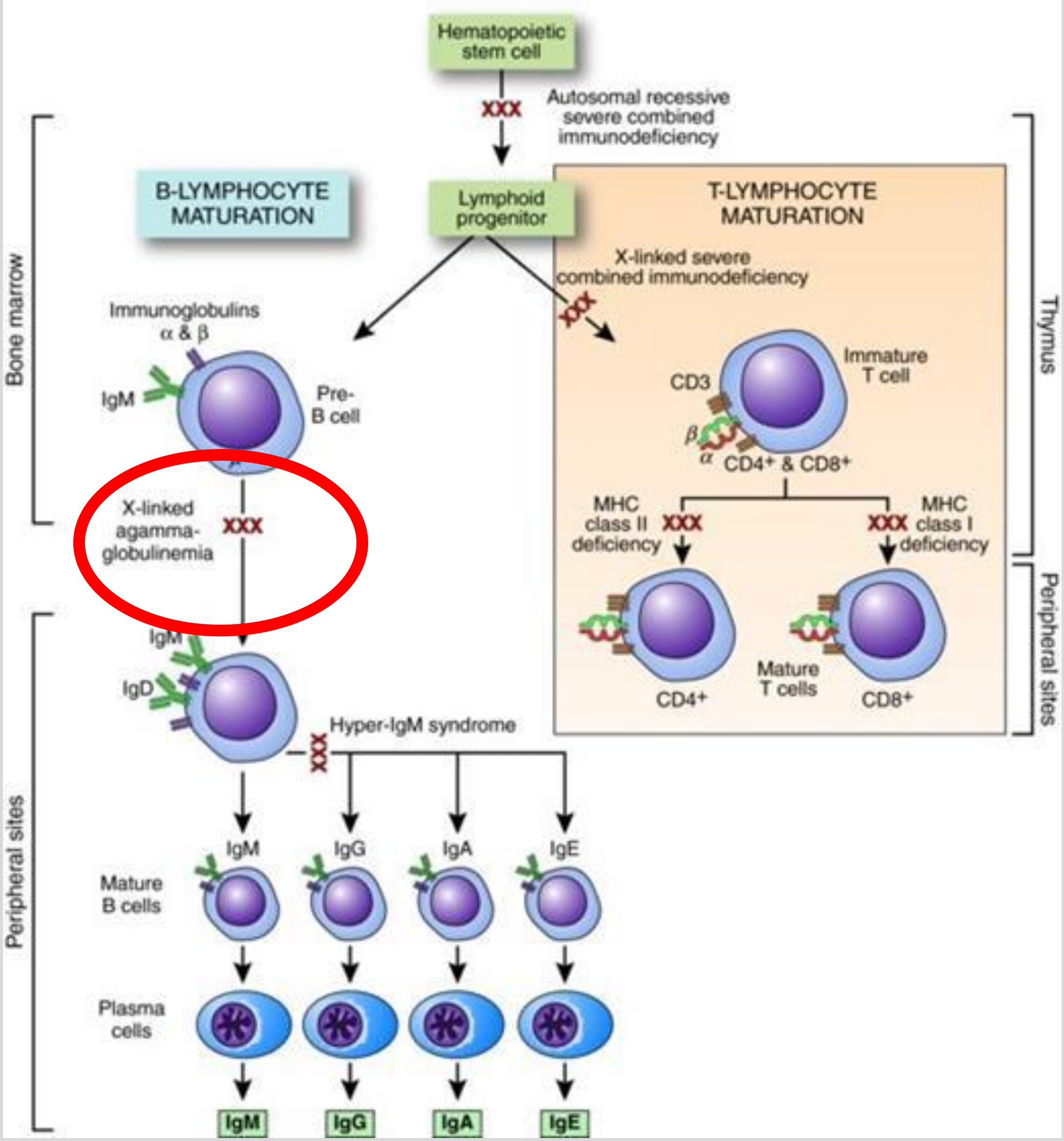




B cell disorder: Bruton X linked agammalobulinemia

- X linked recessive (affecting Boys)
- Defect in BTK gene (tyrosine kinase) which makes light chains in antibodies
- Don't get maturation of B cells
- Absent B cells in peripheral blood!! (CD19+ cells LOW)
  - Trapped in bone marrow
  - Absent lymph nodes/tonsils
  - Primary follicles and germinal centers are absent
- Immunoglobulins all decreased – present after 6 months of age – thanks to mom
- Recurrent bacterial (H flu, Strep Pneumo so recurrent otitis media, sinusitis, pneumonias)
- Enteroviral infections
- Live vaccines contraindicated
- Treatment: lifelong Immunoglobin replacement therapy



Other combined: Wiskott-Aldrich syndrome

- X linked recessive
- Mutation in *WAS* gene
    - Affects leukocytes (B and T cell function) + platelets
    - Unable to reorganize actin cytoskeleton
    - Important in antigen presentation
- Presents with recurrent pyogenic infections (sinus, ear, pneumonia)
- Thrombocytopenia (fewer, smaller PLTs) - bleeding common cause of death
- Eczema
- Increased risk of autoimmune disease and malignancy
- Decreased to normal IgG, IgM levels, increased IgE and IgA levels
- Avoiding live attenuated vaccines
- Tx: Abx ppx, IV immunoglobulins
- Previously bone marrow transplant after conditioning
    - FDA approved 2025 – gene therapy via viral vector –start making WASp

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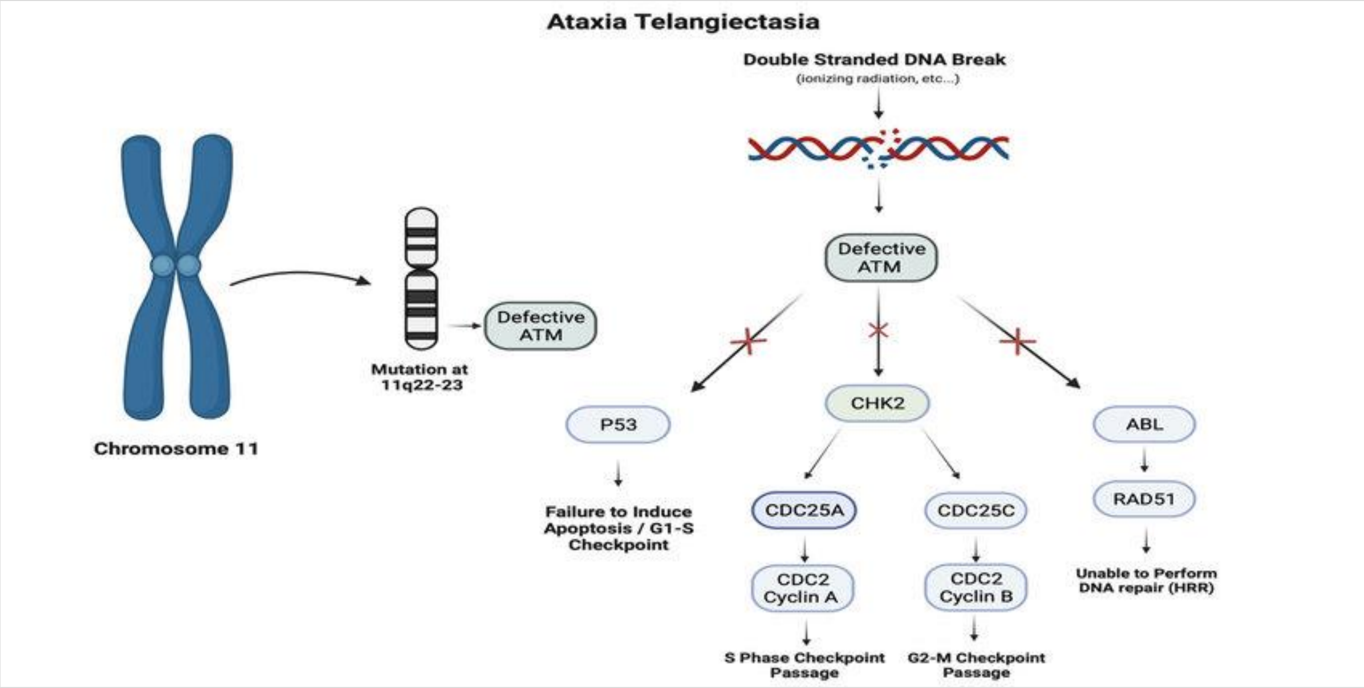
Other combined: Ataxia-telangiectasia (A-T)

- Autosomal Recessive (AR)
- Defects in *ATM* gene (on chromosome 11q22.3
    - Failure to detect DNA manage
    - Failure to stop progression of cell cycle
    - Prone to more mutations
- Onset of symptoms 2-5 years of age, decline may not be noticeable until 4-8 years of age
- Cerebellar Defects (causing ataxia, dysarthria, dysphagia)
- Spider Angiomas (telangiectasia)
    - Cutaneous
    - Ocular
- IgA deficiency
- Increased sensitivity to radiation (limit xray exposure)
- At risk of lymphoma and leukemia
- Increased AFP, IgG and IgE, T cell depression variable
- Lymphopenia, cerebellar atrophy -> leads to uncoordinated movements + falls
- Treatment:
    - Immunoglobulin therapy
    - Ppx Abx



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- Disease process described: SCID (severe combined immunodeficiency)
  - Two flavors – X linked IL-2R mutation or AR - ADA deficient
  - This is a male patient however all three cells (B,T,NK) are decreased – which goes along more with ADA deficiency
  - Chronic diarrhea, Failure to thrive, recurrent viral, fungal, bacterial, protozoal infections
  - NBS would show very low T cells – or TERCs